DESCENT OF FINITE PRESENTATION OF OBJECTS AND MORPHISMS ALONG LIMITS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Author omission kept verbatim: 'Increasing i further (details omitted) we may assume these isomorphisms satisfy the cocycle condition'.

Editorial note. Author omission kept verbatim: 'we omit some of the verifications' at the glueing step.

Editorial note. Both omissions are verification-level; the load-bearing glueing lemma (schemes-lemma-glue) is already proved and cited.

Allowed standard prerequisites (editorial).

- Mor-vs-colimit characterization. Limits, Lemma lemma-descend-finite-presentation and Proposition proposition-characterize-locally-finite-presentation
- Glueing lemma. Schemes, Lemma schemes-lemma-glue

Source reference. Original reference 03GI: sources/stacks-chow/tags.txt, source lines 4529–4529. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05F3: sources/stacks-chow/tags.txt, source lines 7012–7012. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JB: sources/stacks-chow/tags.txt, source lines 2008–2008. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *For a scheme T denote $\mathcal{C}_T$ the full subcategory of schemes W over T such that W is quasi-compact and quasi-separated and such that the structure morphism $W \to T$ is locally of finite presentation. Let $S = \lim S_i$ be a directed limit of schemes with affine transition morphisms. Then there is an equivalence of categories*

$$\operatorname{colim} \mathcal{C}_{S_i} \longrightarrow \mathcal{C}_S$$

given by the base change functors.

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Proof. Fully faithfulness. Suppose we have $i \in I$ and objects X_i, Y_i of $\mathcal{C}_{S_i}$. Denote $X = X_i \times_{S_i} S$ and $Y = Y_i \times_{S_i} S$. Suppose given a morphism $f : X \to Y$ over S. We can choose a finite affine open covering $Y_i = V_{i,1} \cup \ldots \cup V_{i,m}$ such that $V_{i,j} \to Y_i \to S_i$ maps into an affine open $W_{i,j}$ of S_i. Denote $Y = V_1 \cup \ldots \cup V_m$ the induced affine open covering of Y. Since $f : X \to Y$ is quasi-compact (Schemes, Lemma 03GI) after increasing i we may assume that there is a finite open covering $X_i = U_{i,1} \cup \ldots \cup U_{i,m}$ by quasi-compact opens such that the inverse image of $U_{i,j}$ in Y is $f^{-1}(V_j)$, see Lemma 01Z4. By Lemma 01ZM applied to $f|_{f^{-1}(V_j)}$ over W_j we may assume, after increasing i, that there is a morphism $f_{i,j} : V_{i,j} \to U_{i,j}$ over S whose base change to S is $f|_{f^{-1}(V_j)}$. Increasing i more we may assume $f_{i,j}$ and $f_{i,j'}$ agree on the quasi-compact open $U_{i,j} \cap U_{i,j'}$. Then we can glue these morphisms to get the desired morphism $f_i : X_i \to Y_i$. This morphism is unique (up to increasing i) because this is true for the morphisms $f_{i,j}$.

To show that the functor is essentially surjective we argue in exactly the same way. Namely, suppose that X is an object of $\mathcal{C}_S$. Pick $i \in I$. We can choose a finite affine open covering $X = U_1 \cup \ldots \cup U_m$ such that $U_j \to X \to S \to S_i$ factors through an affine open $W_{i,j} \subset S_i$. Set $W_j = W_{i,j} \times_{S_i} S$. This is an affine open of S. By Lemma 01ZM, after increasing i, we may assume there exist $U_{i,j} \to W_{i,j}$ of finite presentation whose base change to W_j is U_j. After increasing i we may assume there exist quasi-compact opens $U_{i,j,j'} \subset U_{i,j}$ whose base changes to S are equal to $U_j \cap U_{j'}$. Claim: after increasing i we may assume the image of the morphism $U_{i,j,j'} \to U_{i,j} \to W_{i,j}$ ends up in $W_{i,j} \cap W_{i,j'}$. Namely, because the complement of $W_{i,j} \cap W_{i,j'}$ is closed in the affine scheme $W_{i,j}$ it is affine. Since $U_j \cap U_{j'} = \lim U_{i,j,j'}$ does map into $W_{i,j} \cap W_{i,j'}$ we can apply Lemma 05F3 to get the claim. Thus we can view both

$$U_{i,j,j'} \quad \text{and} \quad U_{i,j',j}$$

as schemes over $W_{i,j'}$ whose base changes to $W_{j'}$ recover $U_j \cap U_{j'}$. Hence after increasing i, using Lemma 01ZM, we may assume there are isomorphisms $U_{i,j,j'} \to U_{i,j',j}$ over $W_{i,j'}$ and hence over S_i. Increasing i further (details omitted) we may assume these isomorphisms satisfy the cocycle condition mentioned in Schemes, Section 01JA. Applying Schemes, Lemma 01JB we obtain an object X_i of $\mathcal{C}_{S_i}$ whose base change to S is isomorphic to X; we omit some of the verifications. □